

SAINT IGNATIUS' COLLEGE

RIVERVIEW



YEAR 12

EXTENSION TWO MATHEMATICS

ASSESSMENT TASK 1

November 2007

Time allowed: one hour

Instructions to students

- All questions may be attempted.
- All necessary working should be shown in every question.
- Marks for each part in a question shown on the paper.
- Full marks may not be awarded for careless or badly arranged work.
- Board approved calculators and templates may be used.
- The answers to the two questions in this paper are to be returned in separate booklets clearly marked QUESTION 1 and QUESTION 2 on the front cover of the booklet.
- **Write your name on the front cover of each booklet.**

Question 1 (24 marks) Use a SEPARATE writing booklet

Marks

(a) Evaluate:

(i) $|5-2i|$ 1

(ii) $\arg(-3+3i)$ 1

(b) Let $z=2+i$ and $w=1-i$.
Find in the form $x+iy$,

(i) $3z+iw$ 1

(ii) $z\bar{w}$ 2

(iii) $\frac{5}{z}$ 2

(c) (i) Find all pairs of integers a and b such that $(a+ib)^2=8+6i$. 2

(ii) Hence solve: $z^2+2z(1+2i)-(11+2i)=0$. 3

(d) z is a complex number. Show that $z+\frac{|z|^2}{z}$ is real. 2

Question 1 continues on page 2

- (e) (i) If $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$, find z^6 . 1
- (ii) List and also plot on an Argand diagram, all complex numbers that are the solutions of $z^6 = 1$. Answers can be left in modulus-argument form. 2
- (f) (i) Express $z_1 = -1 + i$ and $z_2 = 1 - \sqrt{3}i$ in modulus-argument form. 4
- (ii) Find $z_1 z_2$. 1
- (iii) Hence find the exact value of $\sin \frac{5\pi}{12}$. 2

Question 2 (14 marks) Use a SEPARATE writing booklet

Marks

(a) On separate Argand diagrams, sketch the locus of z described by each of the following conditions:

(i) $|z| \leq 3$ and $0 \leq \arg z \leq \frac{\pi}{3}$ 2

(ii) $2|z| = z + \bar{z} + 4$ 3

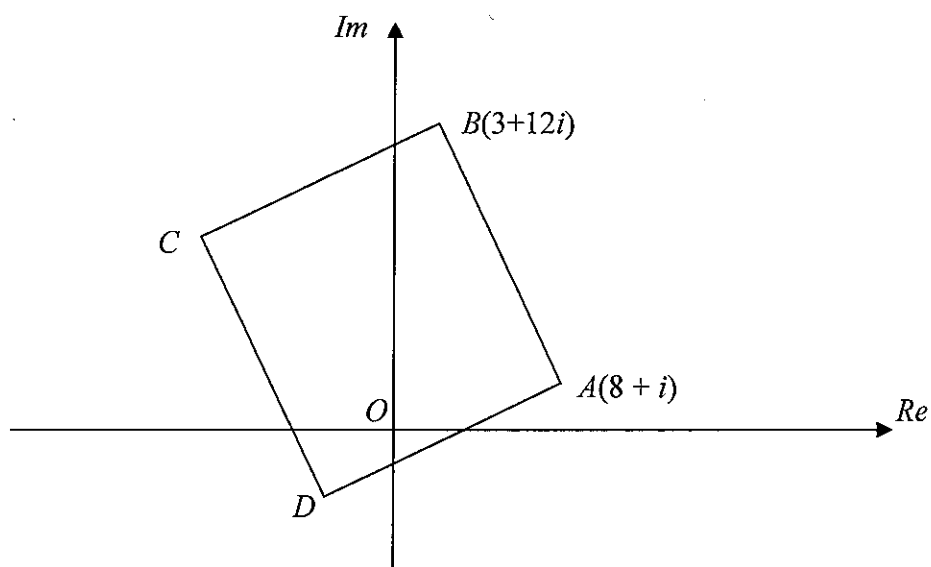
(b) The locus of the complex number z is defined by the equation
 $\arg(z+1) = \frac{\pi}{4}$.

(i) Sketch the locus of z . 1

(ii) Find the least value of $|z|$. 2

Question 2 continues on page 4

(c)



The diagram above shows a square $ABCD$ in the complex plane. The vertices A and B represent the complex numbers $(8 + i)$ and $(3 + 12i)$ respectively. Find the complex numbers represented by:

- | | | |
|------|-------------------|---|
| (i) | the vector AB , | 1 |
| (ii) | the vertex D . | 2 |

- (d) If $w = \frac{1+z}{1-z}$ and $|z|=1$ where w and z are complex numbers, determine the locus of w . 3

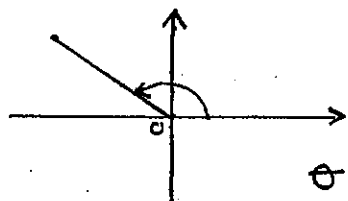
End of Assessment task

Question 1

$$(a) \ i) \ |5-2i| = \sqrt{5^2+2^2} \\ = \sqrt{29}$$

✓

ii)



Let θ be the
argument of $-3+3i$.
 θ is an obtuse angle
 $\tan \theta = -1$

$$\theta = \frac{3\pi}{4}$$

✓

$$b) \ z = 2+i, \ w = 1-i$$

$$i) \ 3z + iw = 3(2+i) + i(1-i) \\ = 6 + 3i + i + 1 \\ = 7 + 4i$$

✓

$$ii) \ z\bar{w} = (2+i)(1+i) \\ = 2 + 3i - 1 \\ = 1 + 3i$$

✓

✓

$$iii) \ \frac{5}{z} = \frac{5}{2+i} \times \frac{2-i}{2-i} \\ = \frac{10-5i}{4+1} \\ = \frac{10-5i}{5} \\ = 2-i$$

✓

✓

$$c) i) (a+ib)^2 = 8+6i$$

$$a^2 + 2abi - b^2 = 8 + 6i$$

$$(a^2 - b^2) + 2abi = 8 + 6i$$

Equating real and imaginary parts,

$$a^2 - b^2 = 8 \dots \textcircled{1}$$

$$2ab = 6 \dots \textcircled{2}$$

Solving $\textcircled{1}$ and $\textcircled{2}$ simultaneously,

$$ab = 3$$

$$a = \frac{3}{b}$$

$$\frac{9}{b^2} - b^2 = 8$$

$$9 - b^4 = 8b^2$$

$$b^4 + 8b^2 - 9 = 0$$

$$(b^2 + 9)(b^2 - 1) = 0$$

$$b^2 = -9 \quad \text{or} \quad b^2 = 1$$

no real sol'n

$$b = \pm 1$$

$$\text{when } b=1, a=3$$

$$\text{when } b=-1, a=-3$$

$\therefore$ solution is $a=3, b=1$ or $a=-3, b=-1$.

$$ii) z = \frac{-2(1+2i) \pm \sqrt{4(1+2i)^2 + 4(1+2i)}}{2}$$

$$= \frac{-2(1+2i) \pm \sqrt{4(1+4i-4) + 44 + 8i}}{2}$$

$$= \frac{-2(1+2i) \pm \sqrt{-12 + 16i + 44 + 8i}}{2}$$

$$= \frac{-2(1+2i) \pm \sqrt{32+24i}}{2}$$

$$= \frac{-2(1+2i) \pm 2\sqrt{8+6i}}{2}$$

$$= -(1+2i) \pm \sqrt{8+6i}$$

From (i) $\sqrt{8+6i} = \pm(3+i)$

$$z = -1-2i+3+i \quad \text{OR} \quad z = -1-2i-3-i$$

$$= 2-i \quad \quad \quad = -4-3i$$

d) Let $z = a+ib$ $|z| = \sqrt{a^2+b^2}$

$$z + \frac{|z|^2}{z} = a+ib + \frac{a^2+b^2}{a+ib}$$

$$= \frac{(a+ib)^2 + a^2 + b^2}{a+ib}$$

$$= \frac{a^2 + 2abi - b^2 + a^2 + b^2}{a+ib}$$

$$= \frac{2a(a+ib)}{(a+ib)}$$

$$= 2a$$

which is real.

OR could have
used $|z|^2 = z\bar{z}$

$$z + \frac{z\bar{z}}{z}$$

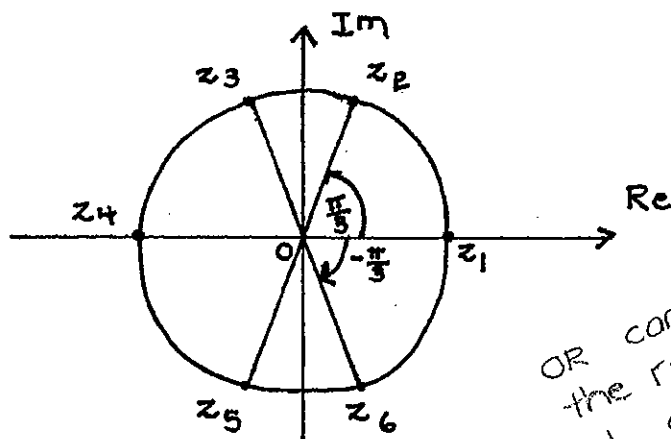
$z + \bar{z}$ is real

e) i) $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$

$$\begin{aligned} z^6 &= \cos\left(6 \times \frac{\pi}{3}\right) + i \sin\left(6 \times \frac{\pi}{3}\right) \\ &= \cos 2\pi + i \sin 2\pi \\ &= 1 + 0i \\ &= 1. \end{aligned}$$

ii) We know that $z = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$ is a solution to $z^6 = 1$. (from part i))
 $z = 1$ and $z = -1$ are also solutions.

The 6 solutions to $z^6 = 1$ are evenly spaced around the unit circle.



$$z_1 = \cos 0 + i \sin 0$$

$$z_2 = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$z_3 = \cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}$$

$$z_4 = \cos \pi + i \sin \pi$$

$$z_5 = \cos\left(-\frac{2\pi}{3}\right) + i \sin\left(-\frac{2\pi}{3}\right) \text{ or } \cos\left(\frac{4\pi}{3}\right) + i \sin\left(\frac{4\pi}{3}\right)$$

$$z_6 = \cos\left(-\frac{\pi}{3}\right) + i \sin\left(-\frac{\pi}{3}\right) \text{ or } \cos\left(\frac{5\pi}{3}\right) + i \sin\left(\frac{5\pi}{3}\right).$$

OR can use the roots of $z^6 = 1$ are given by $z = \cos\left(\frac{2k\pi}{6}\right) + i \sin\left(\frac{2k\pi}{6}\right)$ for $k = 0, 1, 2, 3, 4, 5$.

$$\begin{aligned} \text{f) i) } z_1 &= -1 + i \\ |z_1| &= \sqrt{2} \\ \arg(z_1) &= \frac{3\pi}{4} \end{aligned}$$

$$z_1 = \sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$\begin{aligned} z_2 &= 1 - \sqrt{3}i \\ |z_2| &= \sqrt{4} \\ &= 2 \end{aligned}$$

$$\begin{aligned} \left(\tan^{-1} \sqrt{3} = \frac{\pi}{3} \right) \\ z_2 \text{ lies in 4th} \\ \text{quadrant} \end{aligned}$$

$$\arg(z_2) = -\frac{\pi}{3}$$

$$z_2 = 2 \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right)$$

$$\begin{aligned} \text{ii) } z_1 z_2 &= 2\sqrt{2} \left(\cos \left(\frac{3\pi}{4} - \frac{\pi}{3} \right) + i \sin \left(\frac{3\pi}{4} - \frac{\pi}{3} \right) \right) \\ &= 2\sqrt{2} \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right) \end{aligned}$$

$$\begin{aligned} \text{iii) } z_1 z_2 &= (-1 + i)(1 - \sqrt{3}i) \\ &= -1 + \sqrt{3}i + i + \sqrt{3} \\ &= (\sqrt{3} - 1) + i(\sqrt{3} + 1) \end{aligned}$$

$$\therefore (\sqrt{3} - 1) + i(\sqrt{3} + 1) = 2\sqrt{2} \left(\cos \frac{5\pi}{12} + i \sin \frac{5\pi}{12} \right)$$

Equating imaginary parts,

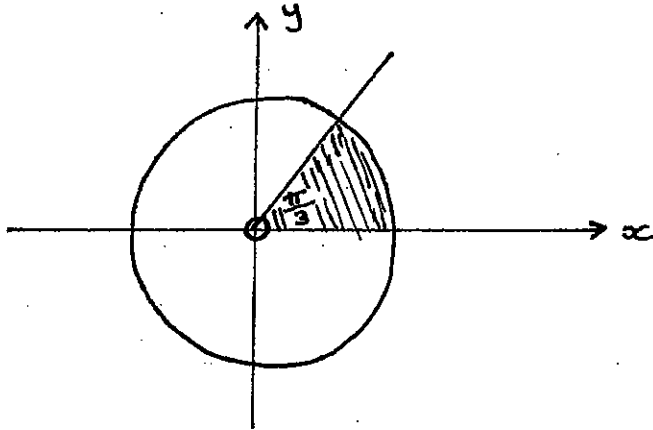
$$\begin{aligned} 2\sqrt{2} \sin \frac{5\pi}{12} &= \sqrt{3} + 1 \\ \sin \frac{5\pi}{12} &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \end{aligned}$$

OR

$$= \frac{\sqrt{6} + \sqrt{2}}{4}$$

Question 2

a) i)



ii) $2|z| = z + \bar{z} + 4$

let $z = x + iy$

$$2\sqrt{x^2 + y^2} = x + iy + x - iy + 4$$

$$2\sqrt{x^2 + y^2} = 2x + 4$$

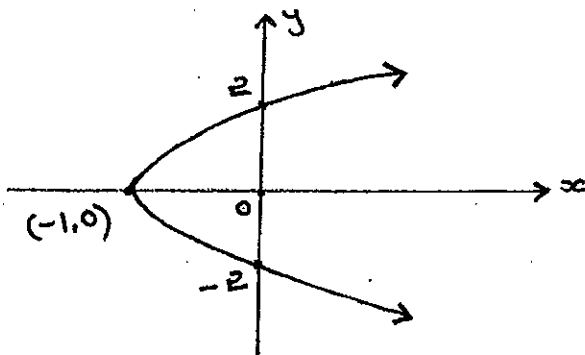
$$\sqrt{x^2 + y^2} = x + 2$$

$$x^2 + y^2 = (x + 2)^2$$

$$x^2 + y^2 = x^2 + 4x + 4$$

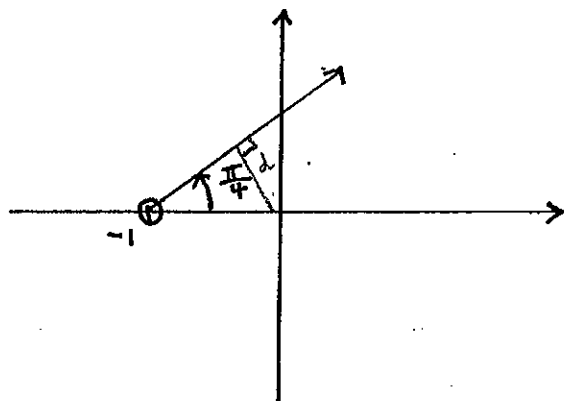
$$y^2 = 4(x + 1)$$

$\therefore$ locus is the parabola with vertex $(-1, 0)$ and focus $(0, 0)$.



b) $\arg(z+1) = \frac{\pi}{4}$

i)



ii) Equation of locus of z .

$$y = x + 1, \quad x > -1$$

Least value of $|z|$ is given by the perpendicular distance of the origin to the line $y = x + 1$.

$$d = \frac{|1|}{\sqrt{1^2 + 1^2}}$$

$$= \frac{1}{\sqrt{2}}$$

$$= \frac{\sqrt{2}}{2}$$

$$\begin{aligned} a &= 1 \\ b &= -1 \\ c &= 1 \\ x_1 &= 0 \\ y_1 &= 0 \end{aligned}$$

or can use trig,

$$\sin \frac{\pi}{4} = \frac{d}{1}$$

$$d = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$\therefore$ least value of $|z|$ is $\frac{\sqrt{2}}{2}$ units.

c) i) $\vec{OA} + \vec{AB} = \vec{OB}$

$$8+i + \vec{AB} = 3+12i$$

$$\vec{AB} = 3+12i - 8-i$$

$$= -5 + 11i$$

ii) $\vec{AD}$ is represented by $(-5+11i)i$
 $= -11 - 5i$ ✓

$$\begin{aligned}\vec{OD} &= \vec{OA} + \vec{AD} \\ \vec{OD} &= (8+i) + (-11-5i) \\ &= -3-4i\end{aligned}$$

so D is represented by $(-3-4i)$. ✓

d) $w = \frac{1+z}{1-z}$ and $|z|=1$.

let $z = x + iy$

OR see
GJA's
solution...

$$\begin{aligned}w &= \frac{1+x+iy}{1-(x+iy)} \\ &= \frac{(1+x)+iy}{(1-x)-iy} \times \frac{(1-x)+iy}{(1-x)+iy} \\ &= \frac{(1+x)(1-x) + (1+x)iy + (1-x)iy - y^2}{(1-x)^2 + y^2}\end{aligned}$$

$$= \frac{1-x^2+iy+\cancel{xyi}+iy-\cancel{xyi}-y^2}{1-2x+x^2+y^2}$$

$$= \frac{1-(x^2+y^2)+2yi}{1-2x+x^2+y^2}$$

given that $|z|=1$, then $x^2+y^2=1$

so $w = \frac{1-1+2yi}{1-2x+1}$

$$= \frac{2yi}{2(1-x)}$$

$$= \frac{y}{(1-x)}i \quad \text{which is purely imaginary}$$

$\therefore$ the locus of w is
the y -axis, or the imaginary axis. ✓

$$W = \frac{1+Z}{1-Z}$$

$$(1-Z)W = 1+Z$$

$$W - ZW = 1 + Z$$

$$W - 1 = ZW + Z$$

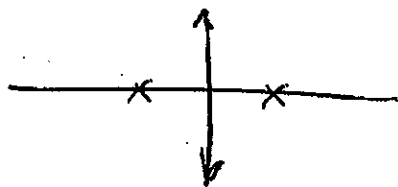
$$W - 1 = Z(W + 1)$$

$$\frac{W-1}{W+1} = Z$$

$$\frac{|W-1|}{|W+1|} = |Z|$$

$$|W-1| = |W+1|$$

$$|W-1| = |W-(-1)|$$



$$\underline{x=0}$$